SAFFIRE-I Operations

Operational Requirements and Protocols

The Saffire experiments are conducted in three consecutive flights of the Cygnus vehicle. Since Cygnus undergoes a destructive re-entry, three Saffire experiment units (Saffire-I, -II, and –III) are constructed and all data must be downlinked. Even with these three flight opportunities, the experiment is very limited in the amount of data and test conditions that can be investigated. In Saffire-I and –III, the sample material is a single large sample (approx. 0.4 m wide by 0.94 m tall) to demonstrate the development and spread of a large-scale low-gravity fire. Once started, the entire burn of each of these samples is recorded, the data compressed, and downlinked. Nine smaller samples are burned on Saffire-II each having a dimension of 5 cm wide x 25 cm long. These are burned sequentially with the camera recording images only from the sample being burned. Once started, these experiments run automatically. Because of limitations in time available for downlinking, a maximum of 20 gigabits of data can be downlinked.

The Saffire-I experiment begins only after Cygnus is unberthed from the ISS. Prior to unberthing, the crew must check that the inlet and outlet ends of the flow duct be clear of any stowage bags being deorbited.

Saffire-I mission operations begin when Cygnus unberths from the ISS. The Cygnus Flight Operations Team (FOT) establishes the vehicle in a lower, circular orbit. Based on this new orbit, ground station contact times are calculated and the experiment start is confirmed and coordinated with the Saffire FOT. From Cygnus undock until experiment start takes about 1 day.

The experiment operations are be conducted in two phases. The first phase consists of turning on power to the experiment avionics, checking the experiment is initialized successfully, starting the experiment run, and recording and compressing the resulting data. The second phase consists of downloading all experiment data via downlink passes at various ground sites. The data is examined for quality and files are retransmitted if necessary. When a complete set of data files has been successfully retrieved, or the timeline has reached a predetermined maximum duration, the experiment avionics power is switched off and the Cygnus vehicle deorbits.

Cygnus burns up on re-entry into the Earth's atmosphere. Consequently, the flight unit is permanently disposed of during this process. No specific disposal processes are expected if the experiment hardware is burned up upon re-entry.